Differential abundance analysis

0.1 Differential abundance analysis

Analysis information:

- Taxonomic level: genus

Taxa that are less prevalent than this (in relabundance) have been agglomerated:

- Detection threshold: 0.1%
- Prevalence threshold: 0.1
- Number of taxonomic groups: 160

BMI is not significantly different between conditions or time points. Hence we can ignore it in the differential abundance testing. In group "B" it seems to be borderline significant ($p<0.05$ before multiple testing adjustment).

Number of significant findings within time points.

- Kruskal test, Benjamini-Hochberg FDR adjustment

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before  after
     0      3
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Number of significant findings within treatments:

- Paired Wilcoxon test, Benjamini-Hochberg FDR adjustment

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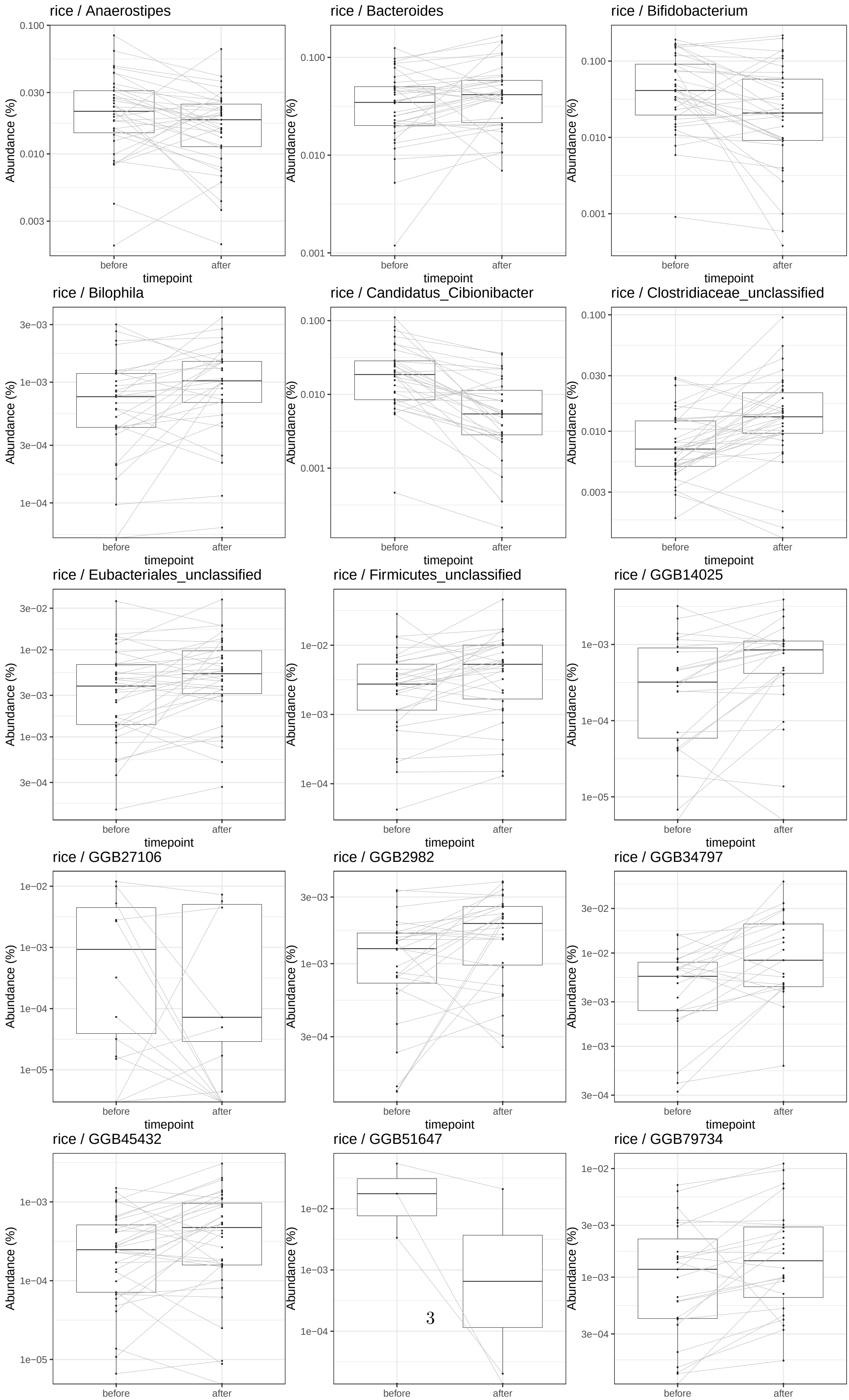
rice  oat
  20    9
```

Let us look more closely to the significant hits. The logFC corresponds to logFC in CLR (tp2-tp1).

Group	Taxon	logFC.clr	FDR	mean.relabundance1	mean.relabundance2
rice	Anaerostipes	-0.0055	0.1771	2.504	1.931
rice	Bacteroides	0.0095	0.1771	3.975	5.003
rice	Bifidobacterium	-0.0171	0.1771	6.265	4.463
rice	Bilophila	0.0003	0.1771	0.074	0.101
rice	Candidatus_Cibionibacter	-0.0147	0.0002	2.336	0.822
rice	Clostridiaceae_unclassified	0.0082	0.0441	0.925	1.771
rice	Eubacteriales_unclassified	0.0017	0.2315	0.546	0.723
rice	Firmicutes_unclassified	0.0025	0.1046	0.379	0.631
rice	GGB14025	0.0002	0.1771	0.040	0.064
rice	GGB27106	-0.0004	0.1771	0.094	0.050
rice	GGB2982	0.0005	0.1771	0.098	0.145
rice	GGB34797	0.0049	0.1771	0.355	0.851
rice	GGB45432	0.0003	0.1771	0.036	0.061
rice	GGB51647	-0.0015	0.2401	0.215	0.060
rice	GGB79734	0.0006	0.1771	0.116	0.178
rice	GGB9559	0.0002	0.1771	0.035	0.056
rice	GGB9760	0.0008	0.1885	0.199	0.277
rice	Gordonibacter	0.0003	0.1885	0.049	0.082
rice	Lachnospiraceae_unclassified	-0.0123	0.1771	3.707	2.400
rice	Lacrimispora	0.0042	0.2043	0.762	1.198

Group	Taxon	logFC.clr	FDR	mean.relabundance1	mean.relabundance2
oat	Butyricicoccus	0.0002	0.1299	0.034	0.051
oat	Candidatus_Cibionibacter	-0.0052	0.0837	1.548	1.014
oat	Eubacteriales_unclassified	-0.0026	0.1894	0.786	0.525
oat	GGB27106	0.0010	0.0837	0.019	0.115
oat	GGB4569	0.0006	0.0240	0.029	0.090
oat	GGB4596	0.0004	0.1299	0.036	0.078
oat	Lawsonibacter	-0.0005	0.1299	0.164	0.111
oat	Parasutterella	0.0006	0.1626	0.076	0.135
oat	Turicibacter	0.0008	0.1626	0.058	0.133


rice / Anaerostipes
rice / Bacteroides
rice / Bifidobacterium
rice / Bilophila
rice / Candidatus_Cibionibacter
rice / Clostridiaceae_unclassified
rice / Eubacteriales_unclassified
rice / Firmicutes_unclassified
rice / GGB14025
rice / GGB27106
rice / GGB2982
rice / GGB34797
rice / GGB45432
rice / GGB51647
rice / GGB79734
Abundance (%)
timepoint
before
after